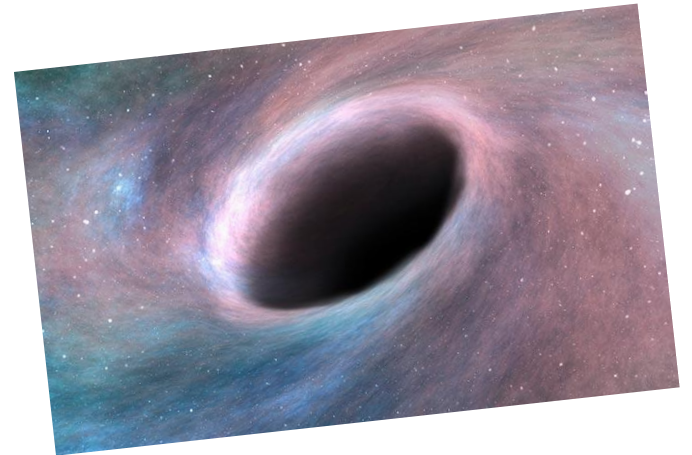


BLACK HOLES



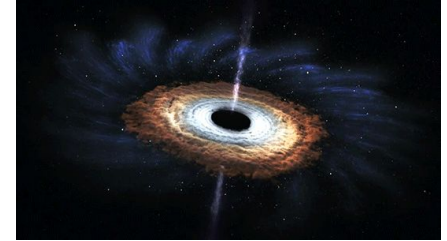
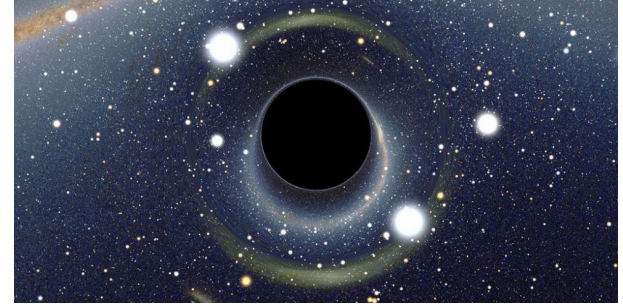
Index

- **WHAT IS a BLACK HOLES**
 - PARTS
- **TYPES OF BLACK HOLES**
- **HOW can SCIENTIST see BLACK HOLES ?**
- **COLLISION OF BLACK HOLES**
- **HISTORY**
 - Important SCIENTIST
- **LOCATION**
-



WHAT IS a BLACK HOLE?

- A PLACE IN SPACE WHERE GRAVITY PULLS SO MUCH THAT EVEN LIGHT CAN NOT GET OUT.
- BECAUSE NO LIGHT CAN GET OUT THEY ARE INVISIBLE TO THE HUMAN EYE.



PARTS OF A BLACK HOLE

THE SINGULARITY

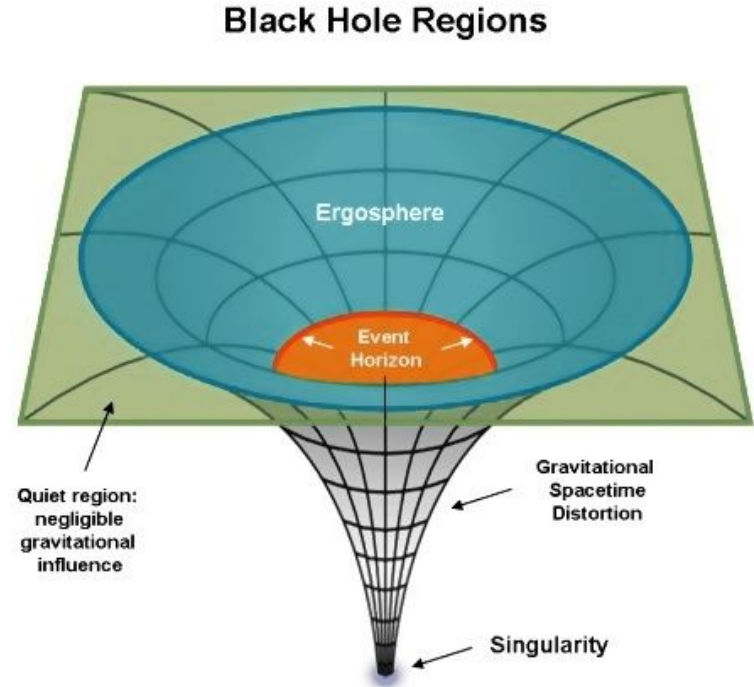
-ALL THE MASS OF THE BLACK HOLE HAS BEEN COMPRESSED DOWN TO NEARLY ZERO VOLUME.

EVENT HORIZON

- THIS IS THE "POINT OF NO RETURN".

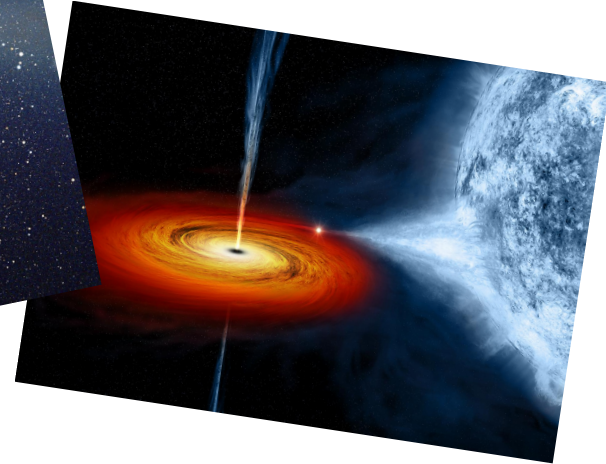
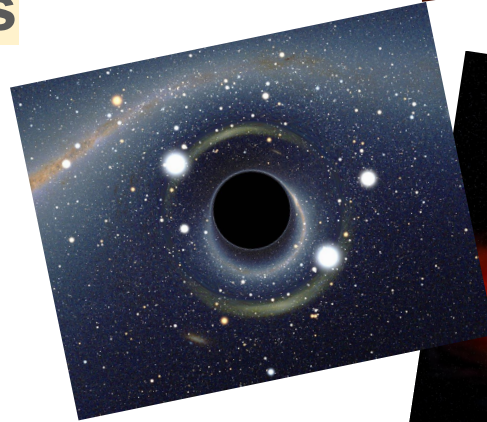
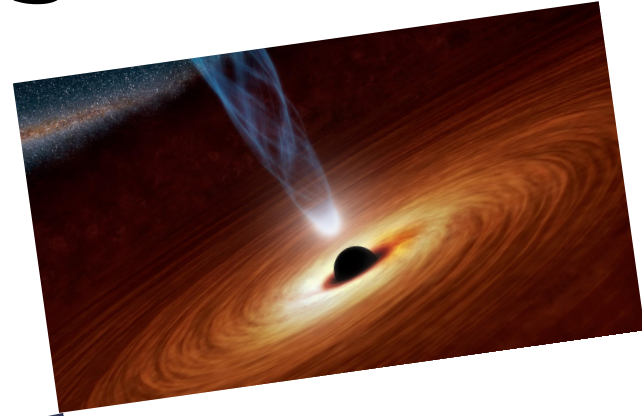
ERGOSPHERE:

IF THE BLACK HOLE IS ROTATING, THEN AS IT SPINS, ITS MASS CAUSES THE SPACE TIME AROUND THE BLACK HOLE TO ROTATE AS WELL.



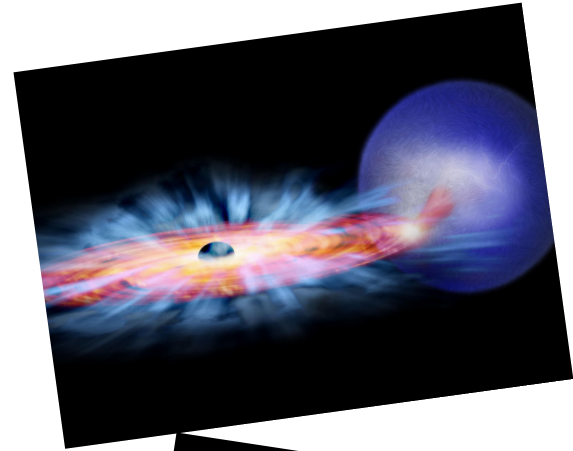
TYPES OF BLACK HOLES

- **Stellar black holes**
- **Supermassive black holes**
- **Miniature black holes**



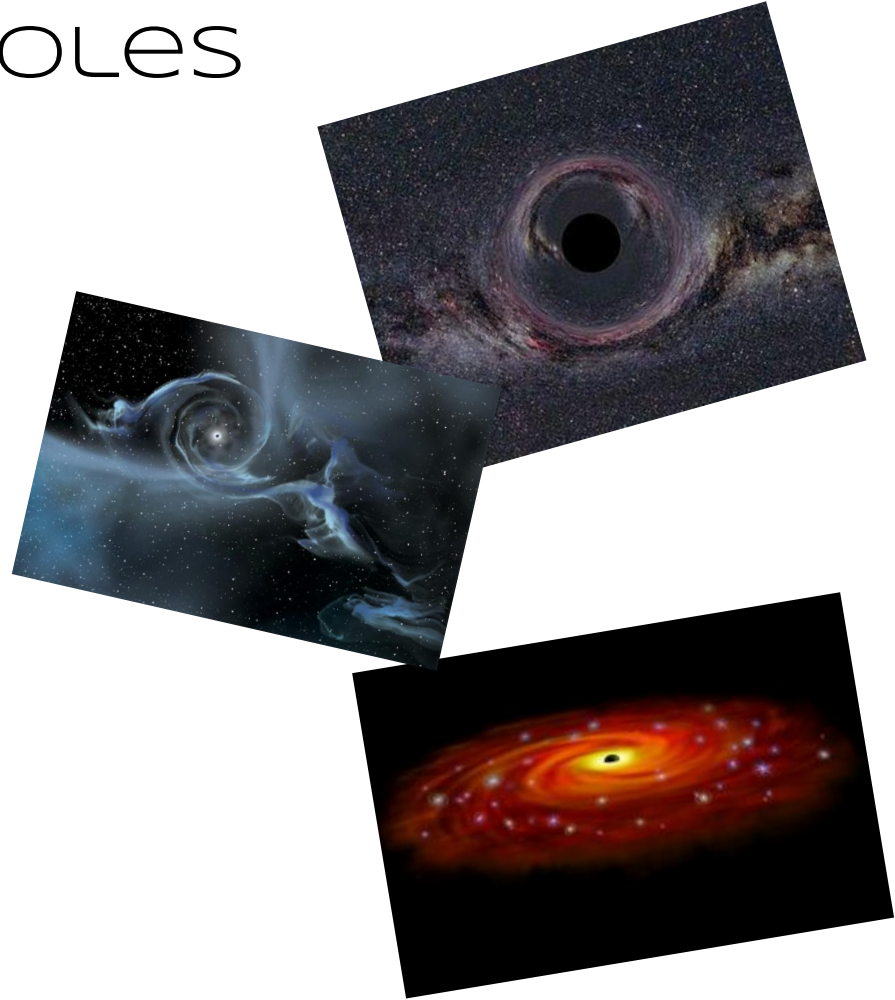
STELLAR BLACK HOLES

- **Can be formed by**
 - Accumulation of neutron star
 - A neutron star merge
 - Collapse supernova
- **Binary system**
- **5 and 10 times the mass of the Sun.**



MINIATURE BLACK HOLES

- 10 billion years ago
- Big Bang
- As small as the width of an atomic particle
- Compress into small point — massive explosion



supermassive BLACK HOLES

- Biggest type of black hole
- 1 million suns together
- Stellar clusters that have collapsed



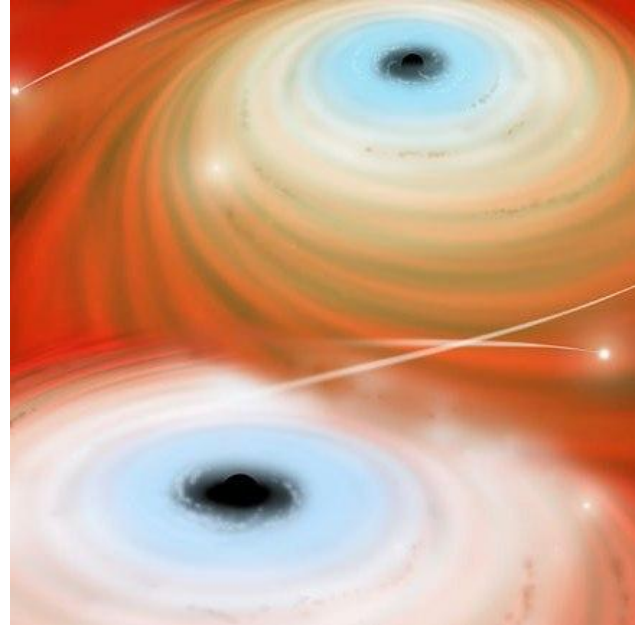
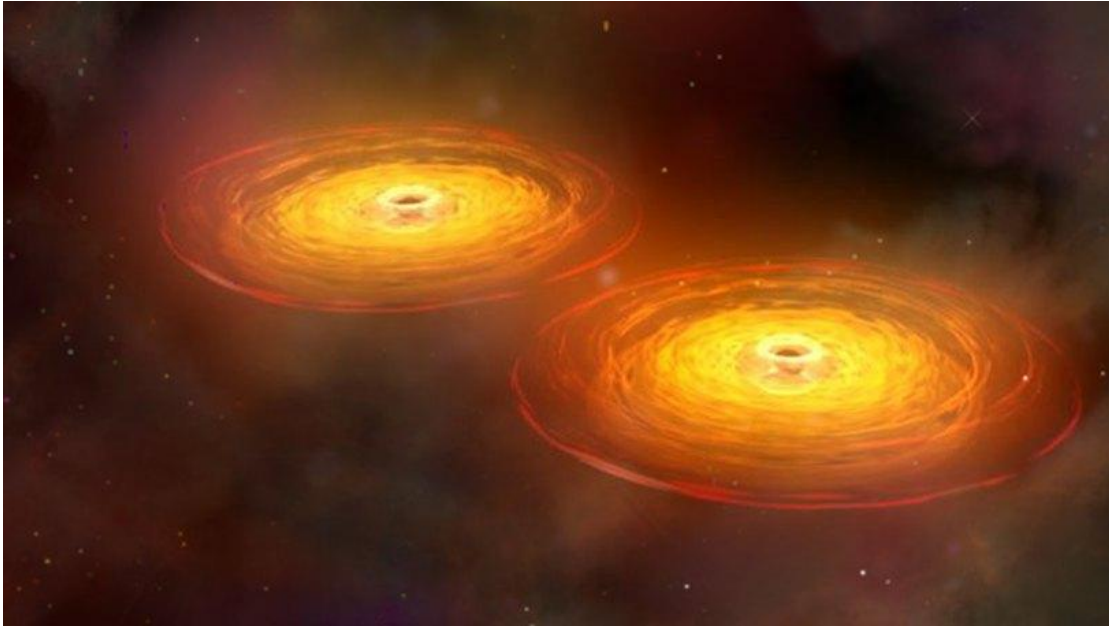
IF BLACK HOLES ARE "BLACK," HOW DO SCIENTISTS KNOW THEY ARE THERE?

SCIENTISTS CAN SEE HOW THE STRONG GRAVITY AFFECTS THE STARS AND GAS AROUND THE BLACK HOLE.

WHEN A BLACK HOLE AND A STAR ARE CLOSE TOGETHER, HIGH-ENERGY LIGHT IS MADE.

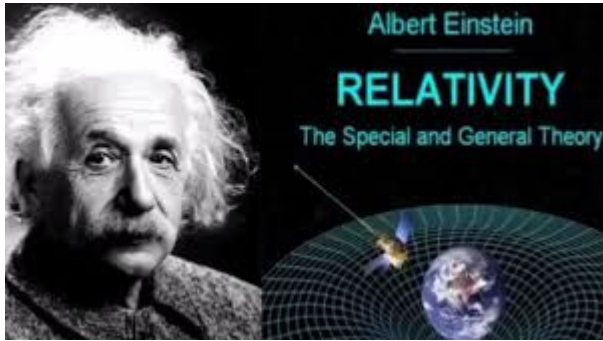


WHAT HAPPEN WHEN BLACK HOLES COLLIDE ?



HISTORY

1915 ALBERT EINSTEIN
PUBLISHED THE THEORY OF
GENERAL RELATIVITY,



1916 KARL
SCHWARZSCHILD
DEFINED BLACK HOLES



1935 SUBRAMANYAN
CHANDRASEKHAR
PUBLISHED THE THEORY
OF WHITE DWARFS.



HISTORY

1964 JOHN WHEELER
COINED THE TERM, "BLACK
HOLE."



1970 STEPHEN
HAWKING DEFINED
MODERN THEORY OF
BLACK HOLES



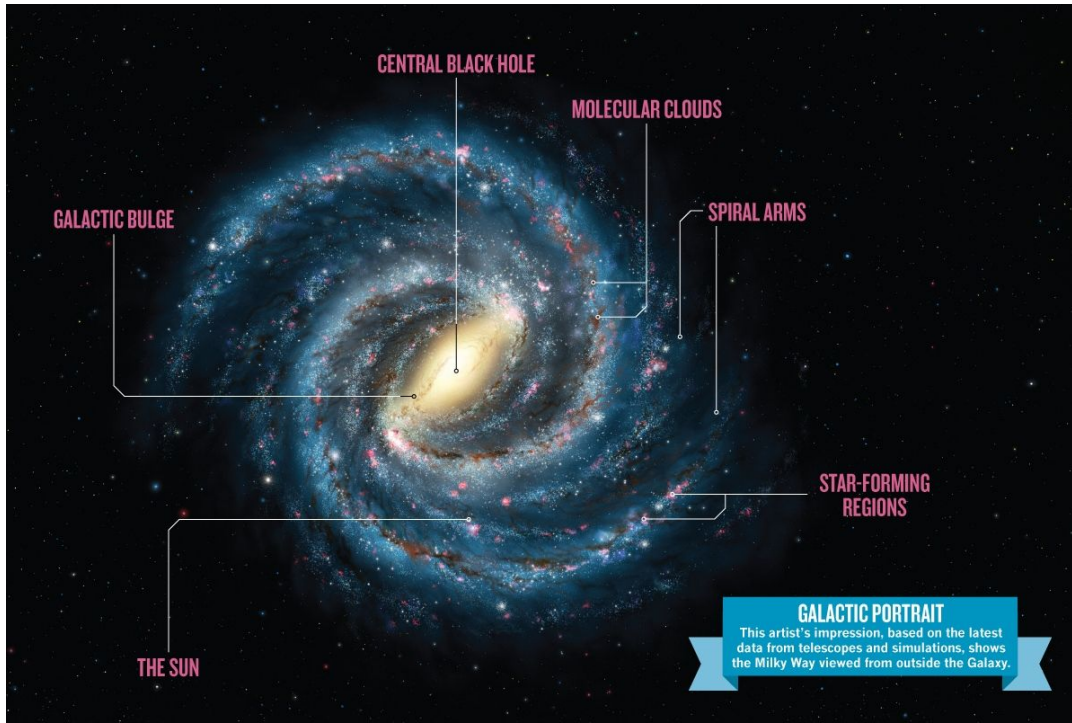
1970 CYGNUS X-1



TODay



SCIENTISTS BELIEVE THERE IS A BLACK HOLE IN THE MIDDLE OF THE MILKY WAY. THEY CALL IT SAGITTARIUS A* AND BELIEVE IT IS 10X THE SIZE OF OUR SOLAR SYSTEM.



THE SUPERMASSIVE BLACK HOLE AT THE CENTER OF THE MILKY WAY GALAXY IS CALLED SAGITTARIUS A.

IT HAS A MASS EQUAL TO ABOUT 4 MILLION SUNS AND WOULD FIT INSIDE A VERY LARGE BALL THAT COULD HOLD A FEW MILLION EARTHS.



UNUSUALLY BRIGHT X-RAY FLARE FROM SAGITTARIUS A*, A BLACK HOLE IN THE CENTER OF THE MILKY WAY GALAXY

STEPHEN HAWKING

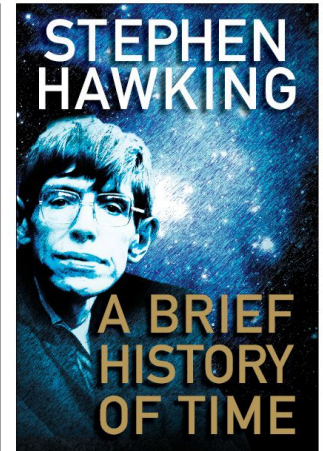
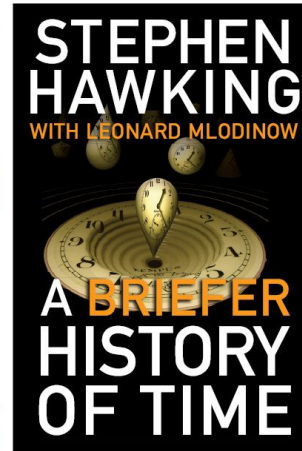
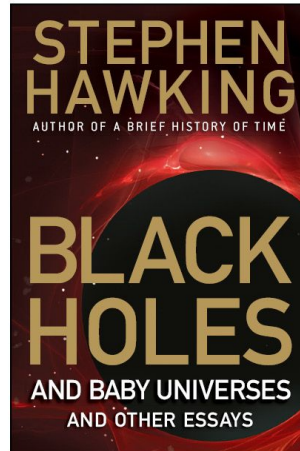


DIAGNOSED WITH LOU GEHRIG'S DISEASE.

STEPHEN HAWKING

HIS THEORY ON BLACK HOLES

BLACK HOLES can only increase, they can never be SPLIT INTO smaller ones. When particles or LIGHT rays pass inside a BLACK HOLE, they never return TO THE rest OF THE universe.



we HOPE you LIKE IT !!

